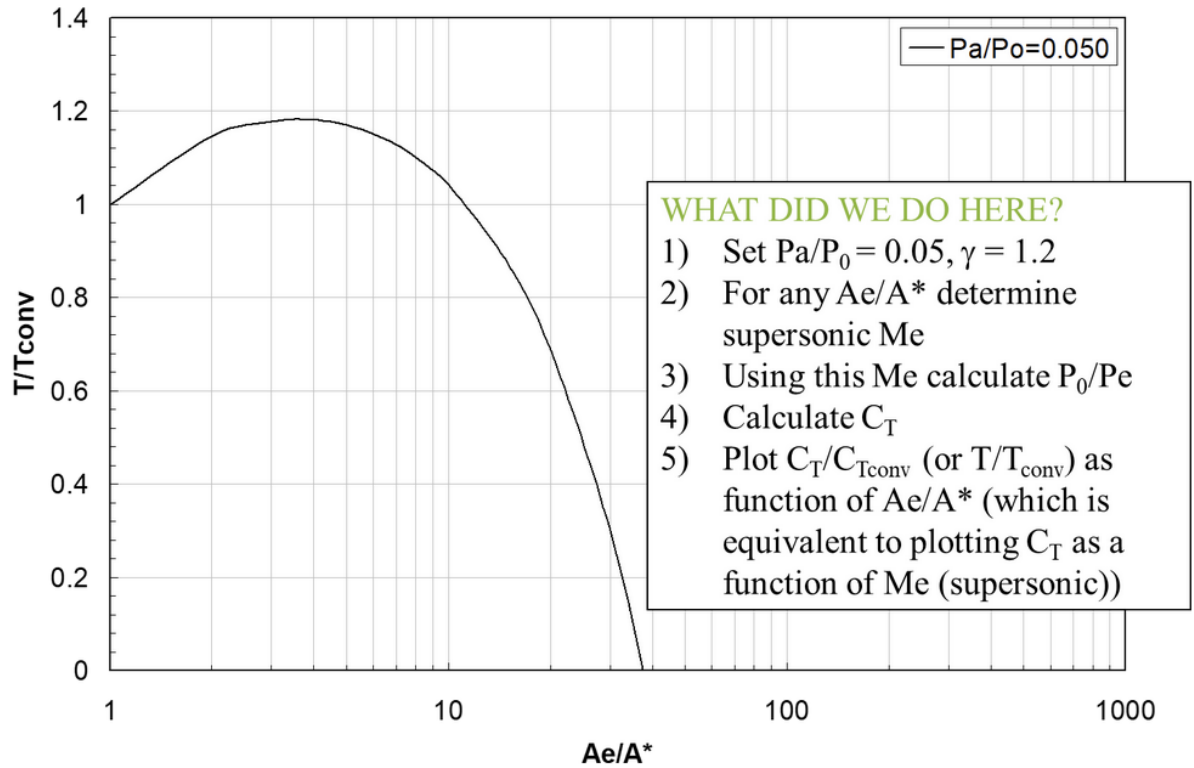


- 1- Using MATLAB or Python, plot this graph of τ / τ_{con} using the directions given in the box and in class. First, use the same values given in the box for P_a/P_0 and γ . Then, keep P_a/P_0 the same and change γ to 1.3, 1.4 and 1.6. Show all four graphs in the same figure with different colors or different line styles.



2. Consider operation of an ideal thrust chamber operating at 11 km altitude ($p_a = 22.7$ kPa). It has the following characteristics:

- Stagnation pressure 1 MPa
- Stagnation temperature 3200 K
- Throat area 0.035 m^2
- Exit area 0.875 m^2
- Propellant gas has $\gamma = 1.25$ and molar mass ($\bar{M}$) = 12

- (a) Verify that the exit velocity of the flow is supersonic.

Hint: Compute pressure ratio at the exit and show that it is greater than critical pressure ratio which is = 16.45.

- (b) Calculate the exit conditions (Mach number, velocity, pressure) of the jet. {Ans.: 4.15, 3889 m/s, 3.19 kPa}
- (c) Determine the mass flow rate and the thrust generated {Ans.: 15.45 kg/s, 43.01 kN}
- (d) From the results of part (c), calculate the characteristic velocity and the thrust coefficient. {Ans.: 2265 m/s, 1.229}